PROBLEM 30

# Particle on a Rotating Quarter Circumference with a Spring

Original statement · Problems.pdf, p. 18

Rotating Frame + Path Variables

A point particle $P$ of mass $m$ is constrained to move on a smooth quarter of circumference as shown in the figure below. We define a moving reference frame $OX_1Y_1Z_1$ that rotates with respect to an inertial reference frame $OXYZ$. The rotation of the moving reference frame is performed along a vertical axis defined as $x_1 = -R$ and $y_1 = 0$; the angular velocity is $\boldsymbol{\omega} = \omega\mathbf{k} = \omega\mathbf{k}_1$. The particle is attached to a linear spring with the other end attached to the point $B$ of the reference frame $OX_1Y_1Z_1$ with coordinates $(-2R, 0, 0)$. The constant of the spring is $K = \alpha m\omega^2$. The angle between $\overline{OP}$ and the negative side of the $z_1$ axis, $\phi$, is used to define the position of the particle. Neglect the action of gravity on the particle.

Initially the particle is located at $\phi = 0$ and it is set to motion with a relative velocity $-\beta\omega R\,\mathbf{i}_1$. Both $\alpha$ and $\beta$ are positive nondimensional parameters.

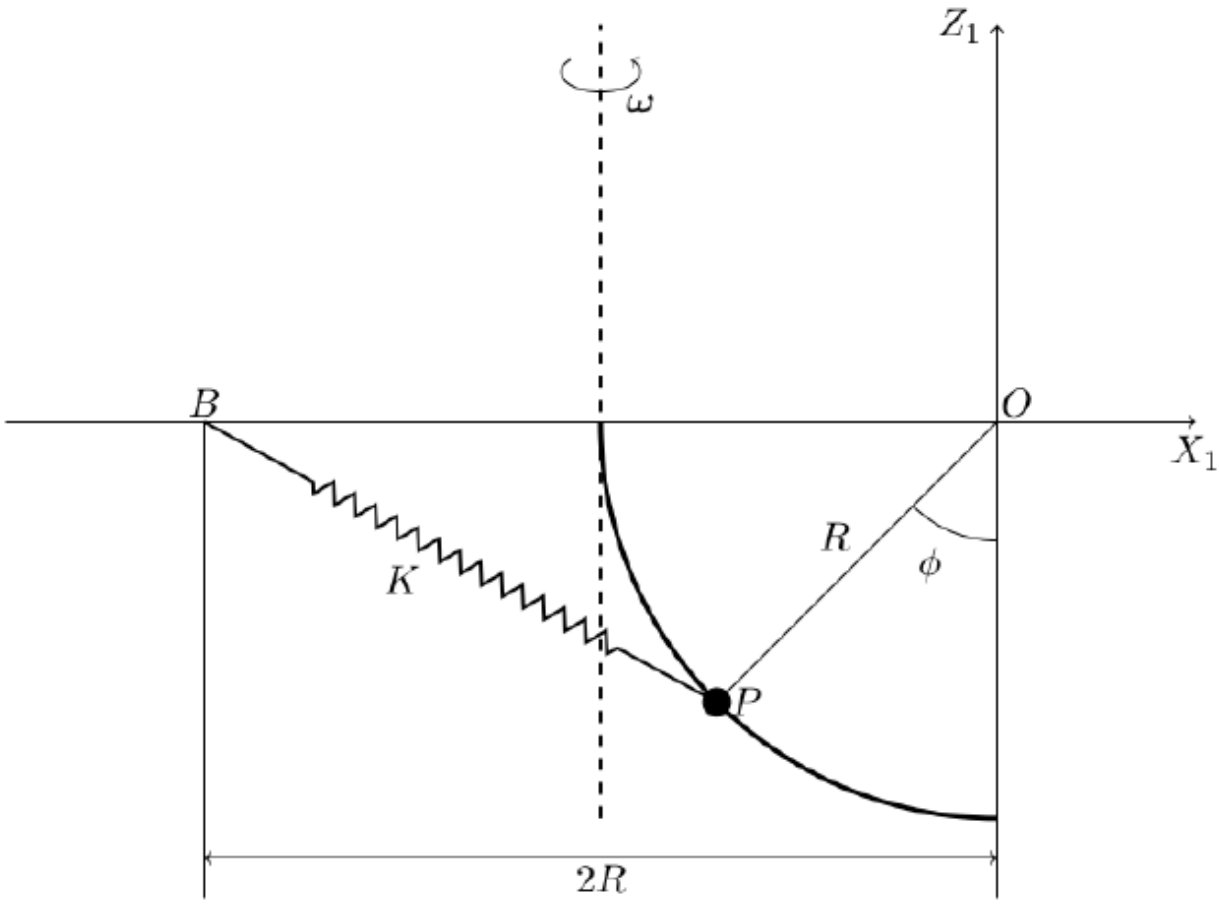



*Figure: Quarter circumference of radius R centred at O in the plane X1Z1, rotation axis x1 = −R (dashed), spring of constant K from B = (−2R, 0, 0) to P, angle φ measured from the negative z1 axis (Problems.pdf, p. 18).*

**(a)** Compute the relative velocity and the relative acceleration of the particle in the path variable reference frame $\{\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b\}$ as functions of the coordinate $\phi$ and their derivatives.

**(b)** Write down the three components of Newton's second law expressed in the path variables reference frame $\{\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b\}$.

**(c)** Express the energy conservation equation in the relative reference frame. Discuss if this equation is equivalent to one of the above ones.

Let us now consider a different problem. Depending on the values of the parameters, we may find the equilibrium of the forces at different positions.

**(d)**

Obtain the value of $\alpha$ to have an equilibrium position when the particle is positioned at $\phi$ equal to $\pi/6$.

For this value of $\alpha$, the particle is again located at $\phi = 0$ and is set to motion with the relative velocity defined before.

**(e)** Describe the possible types of motion for the different values of $\beta$. In particular, obtain the critical value of $\beta$ that makes the particle reach the position $\phi = \pi/2$ and the reaction force of the curve to the particle at that instant.

**(f)** For a value of $\beta$ higher than the critical one, the particle leaves the arc of circumference and its motion is not anymore constrained. Under these conditions compute the maximum height reached by the particle, $z_{max}$, as a function of $\beta$. Note that the particle is still under the action of the spring.

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**Worked Solution (4-Phase Methodology)**

PHASE 1

### Physical Statement, Hypotheses & Parameters

- **System:** a particle $P$ of mass $m$ slides on a smooth quarter circumference of radius $R$ and centre $O$, drawn in the vertical plane $X_1Z_1$ of a frame that rotates with constant angular velocity $\omega\mathbf{k}_1$ about the vertical axis $x_1 = -R, y_1 = 0$. A spring of constant $K = \alpha m\omega^2$ joins $P$ to the point $B = (-2R, 0, 0)$ of the rotating frame. Gravity is neglected.
- **Why a rotating frame.** The wire and the spring anchor $B$ are fixed in $OX_1Y_1Z_1$, so the constraint is a fixed circle there and the position of $P$ is described by one angle. The price is the inertia force of Notes Eqs. 4.62–4.63, which is easy to compute because $\boldsymbol{\omega}_{10}$ is constant. In the inertial frame the wire itself moves, and the constraint would be harder to write.
- **Hypotheses:**
  1. $OXYZ$ is inertial; the frame rotates uniformly, $\boldsymbol{\omega}_{10} = \omega\mathbf{k}_1$ with $\omega$ constant.
  2. The arc is smooth (no friction) and acts as a bilateral constraint (a bead on a wire): the reaction has no tangential component but its normal and binormal components can take

either sign.

3. The spring is linear and, since no natural length is given, of **zero natural length**: $\mathbf{F}_K = -K\,\overrightarrow{BP}$.

4. Gravity is neglected, as the statement says.

- **Degrees of freedom:** on the arc, $P$ has one degree of freedom, the angle $\phi \in [0, \pi/2]$ between $\overline{OP}$ and the negative $z_1$ axis. After leaving the arc in part (f), $P$ is free: three degrees of freedom $(x_1, y_1, z_1)$.
- **Initial conditions:** $\phi(0) = 0$, $\mathbf{v}_1^P(0) = -\beta\omega R\,\mathbf{i}_1$.

| Symbol | Meaning | SI unit |
|---|---|---|
| $m$ | mass of $P$ | kg |
| $R$ | radius of the arc and distance from $O$ to the axis | m |
| $\omega$ | constant angular velocity of the frame | rad/s |
| $K = \alpha m\omega^2$ | spring constant | N/m = kg/s$^2$ |
| $\alpha, \beta$ | positive nondimensional parameters | 1 |
| $\phi$ | angle between $\overline{OP}$ and the $-z_1$ axis | rad |
| $N_n, N_b$ | components of the reaction of the arc | N |

PHASE 2

### Reference Frames, Vector Bases & Rotation Matrices

- **Fixed frame.** The only points that do not move are those of the rotation axis. We take the inertial frame $S_0 = \{Q; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ with origin at the point $Q$ of the axis at the height of $O$ (the point $x_1 = -R$, $y_1 = z_1 = 0$, the upper end of the arc). It plays the role of the statement's $OXYZ$; moving the origin of an inertial frame to another fixed point does not change any velocity or acceleration. $\mathbf{k}_0 = \mathbf{k}$ is vertical.
- **Moving frame.** $S_1 = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$ is the statement's $OX_1Y_1Z_1$, attached to the arc, with $\mathbf{k}_1 = \mathbf{k}_0$. Its origin $O$ is at distance $R$ from the axis, so it moves on a horizontal circle.
- **Change of basis.** $S_1$ turns about $\mathbf{k}_0$ through the angle $\omega t$ (we choose $\mathbf{i}_1 = \mathbf{i}_0$ at $t = 0$). The columns of $[{}_0R_1]$ are the components of $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ in $\mathcal{B}_0$:

$$[{}_0R_1] = \begin{pmatrix} \cos\omega t & -\sin\omega t & 0 \\ \sin\omega t & \cos\omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

- **Checks.** $\det[{}_0R_1] = \cos^2\omega t + \sin^2\omega t = 1$. The columns have unit length ($\cos^2\omega t + \sin^2\omega t = 1$) and the first two are orthogonal ($-\cos\omega t\sin\omega t + \sin\omega t\cos\omega t = 0$), so $[{}_0R_1][{}_0R_1]^T = I$.
- **Angular motion of $S_1$:** $\boldsymbol{\omega}_{10} = \omega\mathbf{k}_1$ and, $\omega$ being constant, $\boldsymbol{\alpha}_{10} = \left.\dfrac{d\boldsymbol{\omega}_{10}}{dt}\right|_0 = \mathbf{0}$.
- **Motion of the origin $O$.** $\mathbf{r}_0^O = \overrightarrow{QO} = R\,\mathbf{i}_1$. By the Coriolis theorem (Notes Eq. 3.8), since $R\,\mathbf{i}_1$ is constant in $S_1$:

$$\mathbf{v}_0^O = \left.\frac{d(R\mathbf{i}_1)}{dt}\right|_1 + \boldsymbol{\omega}_{10}\times R\,\mathbf{i}_1 = \mathbf{0} + \omega R\,(\mathbf{k}_1\times\mathbf{i}_1) = \omega R\,\mathbf{j}_1$$

$$\mathbf{a}_0^O = \left.\frac{d(\omega R\mathbf{j}_1)}{dt}\right|_1 + \boldsymbol{\omega}_{10}\times\omega R\,\mathbf{j}_1 = \omega^2 R\,(\mathbf{k}_1\times\mathbf{j}_1) = -\omega^2 R\,\mathbf{i}_1$$

- **Path (intrinsic) basis.** On the arc, $\mathbf{r}_1^P = -R\sin\phi\,\mathbf{i}_1 - R\cos\phi\,\mathbf{k}_1$. We take the arc length $s = R\phi$ (increasing with $\phi$), the tangent $\mathbf{e}_t$ in the direction of increasing $\phi$, the principal normal $\mathbf{e}_n$ towards the centre of curvature $O$, and $\mathbf{e}_b = \mathbf{e}_t\times\mathbf{e}_n$:

$$\mathbf{e}_t = \frac{1}{R}\frac{d\mathbf{r}_1^P}{d\phi} = -\cos\phi\,\mathbf{i}_1 + \sin\phi\,\mathbf{k}_1, \qquad \mathbf{e}_n = -\frac{\mathbf{r}_1^P}{R} = \sin\phi\,\mathbf{i}_1 + \cos\phi\,\mathbf{k}_1$$

$$\mathbf{e}_b = \mathbf{e}_t\times\mathbf{e}_n = \begin{vmatrix} \mathbf{i}_1 & \mathbf{j}_1 & \mathbf{k}_1 \\ -\cos\phi & 0 & \sin\phi \\ \sin\phi & 0 & \cos\phi \end{vmatrix} = (0-0)\,\mathbf{i}_1 - (-\cos^2\phi - \sin^2\phi)\,\mathbf{j}_1 + (0-0)\,\mathbf{k}_1 = \mathbf{j}_1$$

- The matrix whose columns are $\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b$ in $\mathcal{B}_1$ is

$$[{}_1R_t] = \begin{pmatrix} -\cos\phi & \sin\phi & 0 \\ 0 & 0 & 1 \\ \sin\phi & \cos\phi & 0 \end{pmatrix}, \qquad \det[{}_1R_t] = -\cos\phi\,(0-\cos\phi) - \sin\phi\,(0-\sin\phi) = 1$$

and its columns are unit and mutually orthogonal ($\mathbf{e}_t \cdot \mathbf{e}_n = -\cos\phi\sin\phi + \sin\phi\cos\phi = 0$), so $[{}_1R_t][{}_1R_t]^T = I$.

- **Useful projections:** $\mathbf{i}_1 \cdot \mathbf{e}_t = -\cos\phi$, $\mathbf{i}_1 \cdot \mathbf{e}_n = \sin\phi$, $\mathbf{k}_1 \cdot \mathbf{e}_t = \sin\phi$, $\mathbf{k}_1 \cdot \mathbf{e}_n = \cos\phi$, $\mathbf{j}_1 = \mathbf{e}_b$.

PHASE 3

### Step-by-Step Mathematical Deduction

#### (a) Relative velocity and acceleration in the path basis

**Pedagogical Justification:** The relative velocity and acceleration are the first and second derivatives of $\mathbf{r}_1^P$ taken in $S_1$, where $\mathbf{i}_1, \mathbf{k}_1$ are constant. Only $\phi$ depends on time, so each derivative is a chain rule.

Differentiate $\mathbf{r}_1^P = -R\sin\phi\,\mathbf{i}_1 - R\cos\phi\,\mathbf{k}_1$ with $\dfrac{d}{dt}\sin\phi = \cos\phi\,\dot\phi$ and $\dfrac{d}{dt}\cos\phi = -\sin\phi\,\dot\phi$:

$$\mathbf{v}_1^P = \left.\frac{d\mathbf{r}_1^P}{dt}\right|_1 = -R\cos\phi\,\dot\phi\,\mathbf{i}_1 + R\sin\phi\,\dot\phi\,\mathbf{k}_1 = R\dot\phi\,(-\cos\phi\,\mathbf{i}_1 + \sin\phi\,\mathbf{k}_1)$$

$$\boxed{\mathbf{v}_1^P = R\dot\phi\,\mathbf{e}_t}$$

which is $\dot s\,\mathbf{e}_t$ with $s = R\phi$. For the acceleration use the product rule on $R\dot\phi\,\mathbf{e}_t$, and the chain rule on $\mathbf{e}_t(\phi(t))$:

$$\left.\frac{d\mathbf{e}_t}{dt}\right|_1 = \frac{d\mathbf{e}_t}{d\phi}\,\dot\phi = (\sin\phi\,\mathbf{i}_1 + \cos\phi\,\mathbf{k}_1)\,\dot\phi = \dot\phi\,\mathbf{e}_n$$

$$\mathbf{a}_1^P = \left.\frac{d\mathbf{v}_1^P}{dt}\right|_1 = R\ddot\phi\,\mathbf{e}_t + R\dot\phi\left.\frac{d\mathbf{e}_t}{dt}\right|_1 \implies \boxed{\mathbf{a}_1^P = R\ddot\phi\,\mathbf{e}_t + R\dot\phi^2\,\mathbf{e}_n}$$

The normal part is $\dot{s}^2/\rho$ with radius of curvature $\rho = R$, as expected for a circle. There is no binormal component.

**(b) Newton's second law in the path basis**

> **Pedagogical Justification:** $S_1$ is not inertial, so we use Newton's law with inertia forces, $m\mathbf{a}_1^P = \mathbf{F} + \mathbf{F}_i$ (Notes Eqs. 4.62–4.63, Slides 27–28). This keeps the simple relative acceleration of part (a) and puts the frame's motion into four forces.

**Actual forces.**

- Spring (zero natural length): $\overrightarrow{BP} = \mathbf{r}_1^P - \mathbf{r}_1^B = (2R - R\sin\phi)\,\mathbf{i}_1 - R\cos\phi\,\mathbf{k}_1$, so

$$\mathbf{F}_K = -\alpha m\omega^2 R\,[(2 - \sin\phi)\,\mathbf{i}_1 - \cos\phi\,\mathbf{k}_1]$$

- Reaction of the smooth arc: $\mathbf{N} = N_n\,\mathbf{e}_n + N_b\,\mathbf{e}_b$ (no $\mathbf{e}_t$ component, because the arc is smooth).

**Inertia forces** (Slide 28), with $\boldsymbol{\omega}_{10} = \omega\mathbf{k}_1$, $\boldsymbol{\alpha}_{10} = \mathbf{0}$, $\mathbf{a}_0^O = -\omega^2 R\,\mathbf{i}_1$:

- Force due to the acceleration of the origin: $\mathbf{F}_A = -m\mathbf{a}_0^O = m\omega^2 R\,\mathbf{i}_1$.
- Euler force: $\mathbf{F}_E = -m\boldsymbol{\alpha}_{10} \times \mathbf{r}_1^P = \mathbf{0}$.
- Centrifugal force. First $\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P = \omega\mathbf{k}_1 \times (-R\sin\phi\,\mathbf{i}_1 - R\cos\phi\,\mathbf{k}_1) = -\omega R\sin\phi\,\mathbf{j}_1$ (using $\mathbf{k}_1 \times \mathbf{i}_1 = \mathbf{j}_1$, $\mathbf{k}_1 \times \mathbf{k}_1 = \mathbf{0}$); then $\boldsymbol{\omega}_{10} \times (-\omega R\sin\phi\,\mathbf{j}_1) = -\omega^2 R\sin\phi\,(\mathbf{k}_1 \times \mathbf{j}_1) = \omega^2 R\sin\phi\,\mathbf{i}_1$, so $\mathbf{F}_C = -m\omega^2 R\sin\phi\,\mathbf{i}_1$.
- Coriolis force: $\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P = \omega R\dot{\phi}\,\mathbf{k}_1 \times (-\cos\phi\,\mathbf{i}_1 + \sin\phi\,\mathbf{k}_1) = -\omega R\dot{\phi}\cos\phi\,\mathbf{j}_1$, so $\mathbf{F}_{COR} = -2m\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P = 2m\omega R\dot{\phi}\cos\phi\,\mathbf{j}_1$.

The first and third combine into the centrifugal force about the **true** rotation axis, whose distance to $P$ is $x_1 + R = R(1 - \sin\phi)$ (Slide 30):

$$\mathbf{F}_A + \mathbf{F}_C = m\omega^2 R\,(1 - \sin\phi)\,\mathbf{i}_1$$

**Projections.** With the projections listed in Phase 2:

$$\mathbf{F}_K \cdot \mathbf{e}_t = -\alpha m\omega^2 R\,[(2 - \sin\phi)(-\cos\phi) + (-\cos\phi)(\sin\phi)] = -\alpha m\omega^2 R\,[-2\cos\phi + \sin\phi\cos\phi - \sin\phi\cos\phi] = 2\alpha m\omega^2 R\cos\phi$$

$$\mathbf{F}_K \cdot \mathbf{e}_n = -\alpha m\omega^2 R\left[(2-\sin\phi)\sin\phi - \cos^2\phi\right] = -\alpha m\omega^2 R\left[2\sin\phi - 1\right] = \alpha m\omega^2 R\,(1-2\sin\phi)$$

$$(\mathbf{F}_A + \mathbf{F}_C)\cdot\mathbf{e}_t = -m\omega^2 R\,(1-\sin\phi)\cos\phi, \qquad (\mathbf{F}_A + \mathbf{F}_C)\cdot\mathbf{e}_n = m\omega^2 R\,(1-\sin\phi)\sin\phi$$

and $\mathbf{F}_K$, $\mathbf{F}_A + \mathbf{F}_C$ have no $\mathbf{e}_b = \mathbf{j}_1$ component, while $\mathbf{F}_{COR}$ lies entirely along $\mathbf{e}_b$.

**The three components** of $m\mathbf{a}_1^P = \mathbf{F}_K + \mathbf{N} + \mathbf{F}_A + \mathbf{F}_E + \mathbf{F}_C + \mathbf{F}_{COR}$:

$$\mathbf{e}_t: \quad mR\ddot{\phi} = 2\alpha m\omega^2 R\cos\phi - m\omega^2 R\,(1-\sin\phi)\cos\phi$$

$$\mathbf{e}_n: \quad mR\dot{\phi}^2 = N_n + \alpha m\omega^2 R\,(1-2\sin\phi) + m\omega^2 R\,(1-\sin\phi)\sin\phi$$

$$\mathbf{e}_b: \quad 0 = N_b + 2m\omega R\dot{\phi}\cos\phi$$

The tangential component contains no reaction and is the **equation of motion**; the other two give the **reaction**:

$$\boxed{\ddot{\phi} = \omega^2\cos\phi\;(2\alpha - 1 + \sin\phi)}$$

$$\boxed{N_n = mR\dot{\phi}^2 - \alpha m\omega^2 R\,(1-2\sin\phi) - m\omega^2 R\,\sin\phi\,(1-\sin\phi), \qquad N_b = -2m\omega R\dot{\phi}\cos\phi}$$

The binormal reaction balances the Coriolis force: the arc has to push $P$ sideways to keep it in the rotating plane.

**(c) Energy conservation in the relative frame**

> **Pedagogical Justification:** In $S_1$ (Slide 29) the mechanical energy $E_1^P$ is conserved if every force that does work in $S_1$ has a potential. Here: $\mathbf{N}$ does no work in $S_1$ because the arc is fixed in $S_1$ and $\mathbf{N} \perp \mathbf{v}_1^P = R\dot{\phi}\,\mathbf{e}_t$; $\mathbf{F}_{COR} \perp \mathbf{v}_1^P$ never does work (Slide 31); $\mathbf{F}_E = \mathbf{0}$; the spring is conservative; and $\mathbf{F}_A + \mathbf{F}_C$ is the centrifugal force about an axis fixed in $S_1$ with constant $\omega$, which has the potential of Slide 30.

**Potentials.** Spring: $V_K = \frac{1}{2}K\,|\overrightarrow{BP}|^2$ with

$$|\overrightarrow{BP}|^2 = R^2\left[(2-\sin\phi)^2 + \cos^2\phi\right] = R^2\left[4 - 4\sin\phi + \sin^2\phi + \cos^2\phi\right] = R^2\,(5 - 4\sin\phi)$$

so $V_K = \frac{1}{2}\alpha m\omega^2 R^2\,(5-4\sin\phi)$. Centrifugal (Slide 30), with $r_\perp = x_1 + R = R(1-\sin\phi)$:

$$V_C = -\tfrac{1}{2}m\omega^2 r_\perp^2 = -\tfrac{1}{2}m\omega^2 R^2\,(1-\sin\phi)^2$$

(Check: in Cartesian form $V_C = -\frac{1}{2}m\omega^2(x_1+R)^2$ and $-\partial V_C/\partial x_1 = m\omega^2(x_1+R)$, the $\mathbf{i}_1$ component of $\mathbf{F}_A + \mathbf{F}_C$.)

**Energy equation:**

$$\boxed{E_1^P = \tfrac{1}{2}mR^2\dot{\phi}^2 + \tfrac{1}{2}\alpha m\omega^2 R^2\,(5-4\sin\phi) - \tfrac{1}{2}m\omega^2 R^2\,(1-\sin\phi)^2 = \text{const}}$$

**Initial value.** At $t=0$, $\mathbf{e}_t(0) = -\mathbf{i}_1$, so $\mathbf{v}_1^P(0) = R\dot{\phi}(0)\,(-\mathbf{i}_1) = -\beta\omega R\,\mathbf{i}_1$ gives $\dot{\phi}(0) = \beta\omega$, and $E_1^P(0) = \frac{1}{2}m\omega^2R^2\,(\beta^2 + 5\alpha - 1)$. Equating and dividing by $\frac{1}{2}mR^2$:

$$\dot{\phi}^2 = \omega^2\left[\beta^2 + 5\alpha - 1 - \alpha(5-4\sin\phi) + (1-\sin\phi)^2\right] \;\Longrightarrow\; \boxed{\dot{\phi}^2 = \omega^2\left[\beta^2 + (4\alpha-2)\sin\phi + \sin^2\phi\right]}$$

**Equivalence.** Differentiate $E_1^P$ in time with the chain rule ($\frac{d}{dt}\sin\phi = \cos\phi\,\dot{\phi}$, $\frac{d}{dt}(1-\sin\phi)^2 = 2(1-\sin\phi)(-\cos\phi\,\dot{\phi})$):

$$\frac{dE_1^P}{dt} = mR^2\dot\phi\ddot\phi - 2\alpha m\omega^2 R^2\cos\phi\,\dot\phi + m\omega^2R^2(1-\sin\phi)\cos\phi\,\dot\phi = mR^2\dot\phi\left[\ddot\phi - 2\alpha\omega^2\cos\phi + \omega^2(1-\sin\phi)\cos\phi\right]$$

The bracket is zero exactly when the $\mathbf{e}_t$ equation of part (b) holds. So the energy equation is **not new information**: it is the first integral of the tangential component (the $\mathbf{e}_t$ equation multiplied by $R\dot\phi$ and integrated once). The $\mathbf{e}_n$ and $\mathbf{e}_b$ components are independent of it and are needed to compute the reaction.

**(d) Value of $\alpha$ for equilibrium at $\phi = \pi/6$**

> **Pedagogical Justification:** Relative equilibrium in $S_1$ means $\dot\phi = 0$ and $\ddot\phi = 0$ for all time. The $\mathbf{e}_t$ equation then requires that the tangential force vanish.

$$\ddot\phi = 0 \iff \cos\phi\,(2\alpha - 1 + \sin\phi) = 0$$

The root $\cos\phi = 0$ ($\phi = \pi/2$, the particle on the axis) is an equilibrium for every $\alpha$. Any other equilibrium must satisfy $\sin\phi_e = 1 - 2\alpha$. For $\phi_e = \pi/6$, $\sin\phi_e = \tfrac{1}{2}$:

$$2\alpha - 1 + \tfrac{1}{2} = 0 \implies \boxed{\alpha = \tfrac{1}{4}}$$

At that equilibrium, from part (b) with $\dot\phi = 0$: $N_b = 0$ and $N_n = -\tfrac{1}{4}m\omega^2R\,(1-1) - m\omega^2R\cdot\tfrac{1}{2}\cdot\tfrac{1}{2} = -\tfrac{1}{4}m\omega^2R$, i.e. the arc pulls $P$ away from $O$ with force $\tfrac{1}{4}m\omega^2R$.

**(e) Types of motion for $\alpha = 1/4$, critical $\beta$ and reaction at $\phi = \pi/2$**

> **Pedagogical Justification:** The energy integral of part (c) is a first-order equation $\dot\phi^2 = \omega^2[\beta^2 - W(\phi)]$; motion is possible only where $W(\phi) \le \beta^2$, so the shape of $W$ classifies the motions, as with a one-dimensional potential.

With $\alpha = \tfrac{1}{4}$, $4\alpha - 2 = -1$:

$$\dot\phi^2 = \omega^2\left[\beta^2 - W(\phi)\right], \qquad W(\phi) = \sin\phi\,(1-\sin\phi)$$

On $[0,\pi/2]$: $W(0)=0$, $W(\pi/2)=0$, and $\dfrac{dW}{d\phi}=\cos\phi\,(1-2\sin\phi)$ vanishes inside the interval only at $\sin\phi=\tfrac12$, i.e. $\phi=\pi/6$, where $W$ has its maximum $W(\pi/6)=\tfrac12\cdot\tfrac12=\tfrac14$. $W$ plays the role of a potential barrier, so the equilibrium $\phi=\pi/6$ is **unstable**. Three types of motion follow:

- $\beta>\tfrac12$**:** $\dot\phi^2\geq\omega^2(\beta^2-\tfrac14)>0$ on the whole arc, so $\phi$ increases monotonically, crosses the barrier at $\pi/6$ and reaches $\phi=\pi/2$ with $\dot\phi^2=\omega^2[\beta^2-0]$, i.e. $\dot\phi=\beta\omega$: the relative speed there equals the initial one, $\beta\omega R$. The particle then leaves the arc at its upper end (part f).
- $\beta=\tfrac12$**:** $\beta^2-W=\tfrac14-\sin\phi+\sin^2\phi=(\tfrac12-\sin\phi)^2$, so $\dot\phi=\omega(\tfrac12-\sin\phi)$. The particle approaches the unstable equilibrium $\phi=\pi/6$ asymptotically and never reaches it: near it, $\sin\phi\approx\tfrac12+\tfrac{\sqrt3}{2}(\phi-\tfrac{\pi}{6})$ and $\dot\phi\approx-\tfrac{\sqrt3}{2}\omega\,(\phi-\tfrac{\pi}{6})$, an exponential approach with rate $\tfrac{\sqrt3}{2}\omega$.
- $\beta<\tfrac12$**:** $\dot\phi$ vanishes at the turning point $\phi^*$ with $W(\phi^*)=\beta^2$, i.e. $\sin^2\phi^*-\sin\phi^*+\beta^2=0$; the root reached first is

$$\sin\phi^*=\frac{1-\sqrt{1-4\beta^2}}{2}<\frac12$$

There $\ddot\phi=\omega^2\cos\phi^*\,(\sin\phi^*-\tfrac12)<0$, so the particle turns back, returns to $\phi=0$ with $\dot\phi=-\beta\omega$ (relative velocity $+\beta\omega R\,\mathbf{i}_1$, same speed by energy conservation) and leaves the quarter circumference at its lower end.

**Critical value.** The particle reaches $\phi=\pi/2$ if and only if $\beta$ exceeds

$$\boxed{\beta_c=\tfrac12}$$

**Reaction at $\phi=\pi/2$.** Here $\cos\phi=0$, $\sin\phi=1$, $\dot\phi=\beta\omega$, and $\mathbf{e}_n(\pi/2)=\mathbf{i}_1$. From part (b) with $\alpha=\tfrac14$:

$$N_b=-2m\omega R\,\beta\omega\cdot0=0,\qquad N_n=mR\beta^2\omega^2-\tfrac14 m\omega^2R\,(1-2)-m\omega^2R\cdot1\cdot0=m\omega^2R\left(\beta^2+\tfrac14\right)$$

$$\boxed{\mathbf{N}(\phi=\tfrac{\pi}{2})=m\omega^2R\left(\beta^2+\tfrac14\right)\mathbf{i}_1\ \xrightarrow{\beta\to\beta_c}\ \tfrac12 m\omega^2R\,\mathbf{i}_1}$$

It points towards $O$: it supplies the centripetal force $mR\dot{\phi}^2$ and also balances the spring, which pulls $P$ towards $B$ with $\frac{1}{4}m\omega^2 R$. The centrifugal and Coriolis forces vanish there because $P$ is on the axis and moves parallel to it.

**(f) Maximum height after leaving the arc ($\beta > \beta_c$)**

> **Pedagogical Justification:** Once free, $P$ has three degrees of freedom, and the horizontal motion is coupled through the Coriolis and centrifugal forces. But every inertia force of a frame rotating about the vertical is horizontal, so the vertical component of $m\mathbf{a}_1^P = \mathbf{F} + \mathbf{F}_i$ involves only the spring and decouples.

**Initial state of the free motion** (time $\tau = 0$ at separation): $\phi = \pi/2$ gives $\mathbf{r}_1^P = -R\,\mathbf{i}_1$ (on the axis, $z_1 = 0$), and $\mathbf{v}_1^P = R\dot{\phi}\,\mathbf{e}_t(\pi/2) = \beta\omega R\,\mathbf{k}_1$ (vertical, upward).

**Vertical equation.** For a free position $\mathbf{r}_1^P = x_1\mathbf{i}_1 + y_1\mathbf{j}_1 + z_1\mathbf{k}_1$:

- $\mathbf{k}_1 \cdot \mathbf{F}_K = -K\,\mathbf{k}_1 \cdot [(x_1 + 2R)\mathbf{i}_1 + y_1\mathbf{j}_1 + z_1\mathbf{k}_1] = -\alpha m\omega^2 z_1$;
- $\mathbf{k}_1 \cdot (\mathbf{F}_A + \mathbf{F}_C) = 0$, because $\mathbf{F}_A + \mathbf{F}_C = m\omega^2\,[(x_1 + R)\mathbf{i}_1 + y_1\mathbf{j}_1]$ is horizontal;
- $\mathbf{k}_1 \cdot \mathbf{F}_{COR} = -2m\,\mathbf{k}_1 \cdot (\omega\mathbf{k}_1 \times \mathbf{v}_1^P) = 0$, because $\mathbf{k}_1 \times \mathbf{v}_1^P \perp \mathbf{k}_1$.

Hence, with $\alpha = \frac{1}{4}$,

$$m\ddot{z}_1 = -\alpha m\omega^2 z_1 \implies \ddot{z}_1 + \frac{\omega^2}{4} z_1 = 0$$

a harmonic oscillator of angular frequency $\omega/2$. Its general solution and derivative (chain rule $\frac{d}{d\tau}\sin\frac{\omega\tau}{2} = \frac{\omega}{2}\cos\frac{\omega\tau}{2}$) are

$$z_1 = C_1\cos\frac{\omega\tau}{2} + C_2\sin\frac{\omega\tau}{2}, \qquad \dot{z}_1 = -\frac{\omega}{2}C_1\sin\frac{\omega\tau}{2} + \frac{\omega}{2}C_2\cos\frac{\omega\tau}{2}$$

Initial conditions: $z_1(0) = C_1 = 0$ and $\dot{z}_1(0) = \frac{\omega}{2}C_2 = \beta\omega R$, so $C_2 = 2\beta R$:

$$z_1(\tau) = 2\beta R\,\sin\frac{\omega\tau}{2} \implies \boxed{z_{max} = 2\beta R} \quad \text{reached at } \tau = \pi/\omega$$

Since $\mathbf{k}_1 = \mathbf{k}$, this is also the height in the inertial frame, measured from the horizontal plane through $O$ and $B$. For a general $\alpha$ the same steps give $z_{max} = \beta R/\sqrt{\alpha}$. (Equivalently, the vertical motion conserves $\frac{1}{2}m\dot{z}_1^2 + \frac{1}{2}Kz_1^2$, and $\frac{1}{2}m(\beta\omega R)^2 = \frac{1}{2}\cdot\frac{1}{4}m\omega^2 z_{max}^2$ gives the same

result.) The result assumes that the free particle does not hit the wire again.

PHASE 4

### Physical Interpretation & Dimensional Verification

- **Competition of two forces.** Along the arc the spring pulls $P$ towards $B$ (towards larger $\phi$) with tangential force $2\alpha m\omega^2 R\cos\phi$, while the centrifugal force about the axis pushes it away from the axis (towards smaller $\phi$) with $m\omega^2 R(1-\sin\phi)\cos\phi$. Both scale with $\omega^2$ because $K=\alpha m\omega^2$, so $\omega$ only sets the time scale $1/\omega$ and the dynamics depends on $\alpha$ and $\beta$ alone.
- **Limiting cases.** For $\alpha>\frac{1}{2}$ the factor $2\alpha-1+\sin\phi$ is positive on the whole arc and there is no interior equilibrium. For $\beta\to\beta_c^+$ the time to cross the barrier grows without bound (logarithmically), consistent with the asymptotic motion at $\beta=\beta_c$. As $\phi\to\pi/2$, both the Coriolis force and $N_b$ vanish, as they must for motion parallel to $\boldsymbol{\omega}_{10}$.
- **Inertial check.** Computing $\mathbf{a}_0^P$ by differentiating $\mathbf{r}_0^P=\mathbf{r}_0^O+[{}_0R_1]\mathbf{r}_1^P$ twice gives exactly $\mathbf{a}_0^O+\mathbf{a}_1^P+\boldsymbol{\omega}_{10}\times(\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P)+2\boldsymbol{\omega}_{10}\times\mathbf{v}_1^P$ (Notes Eq. 3.27), and $m\mathbf{a}_0^P=\mathbf{F}_K+\mathbf{N}$ holds with the reactions of part (b) (checked symbolically with sympy).
- **Dimensions.** $[K]=[\alpha m\omega^2]=\mathrm{kg}\cdot\mathrm{s}^{-2}=\mathrm{N/m}$; $[\ddot{\phi}]=[\omega^2]=\mathrm{s}^{-2}$; $[N_n]=[m\omega^2 R]=\mathrm{kg}\cdot\mathrm{m}\cdot\mathrm{s}^{-2}=\mathrm{N}$; $[E_1^P]=[m\omega^2R^2]=\mathrm{J}$; $[z_{max}]=[\beta R]=\mathrm{m}$.

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*Source: Problems.pdf (PDF page 18)*